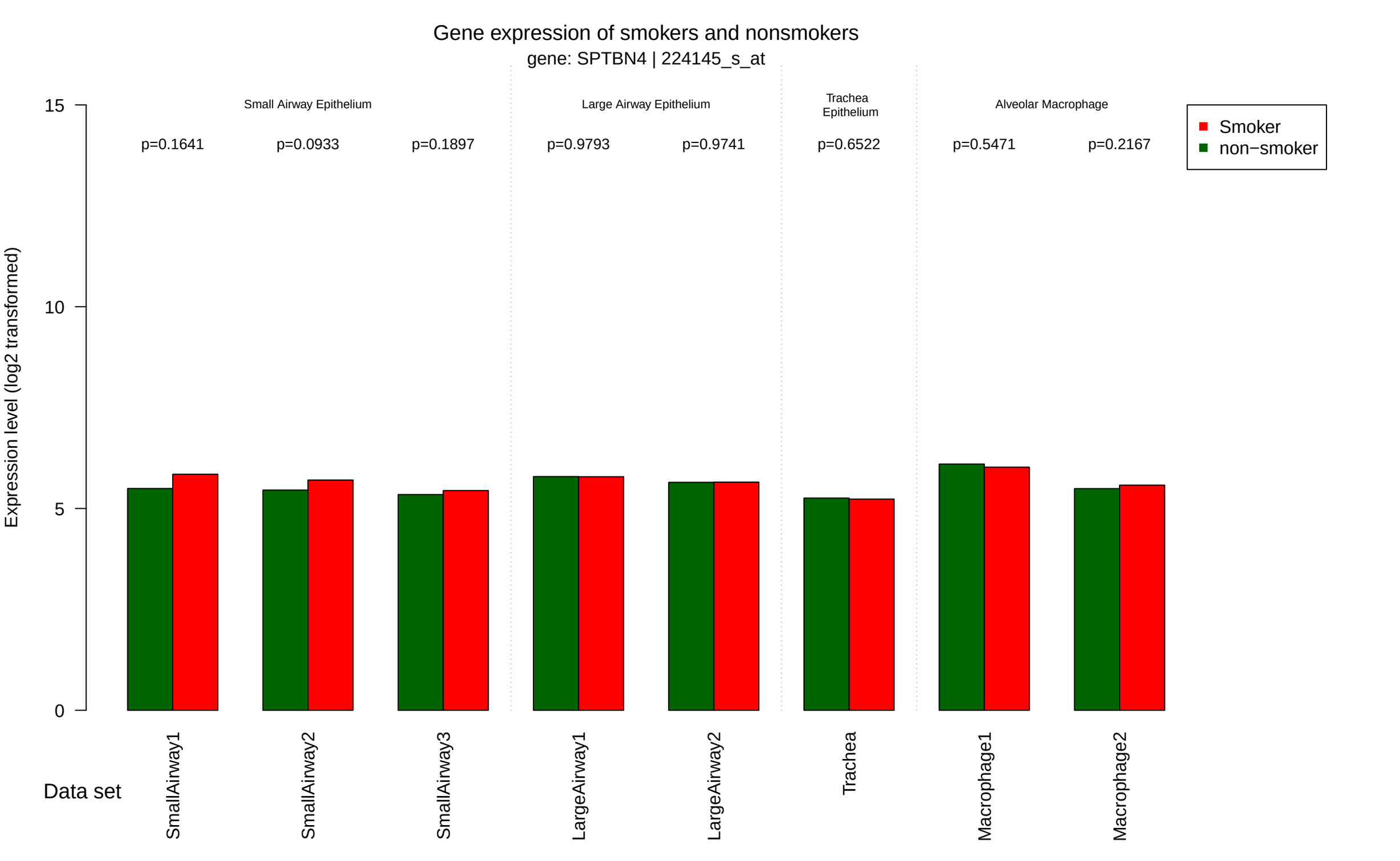
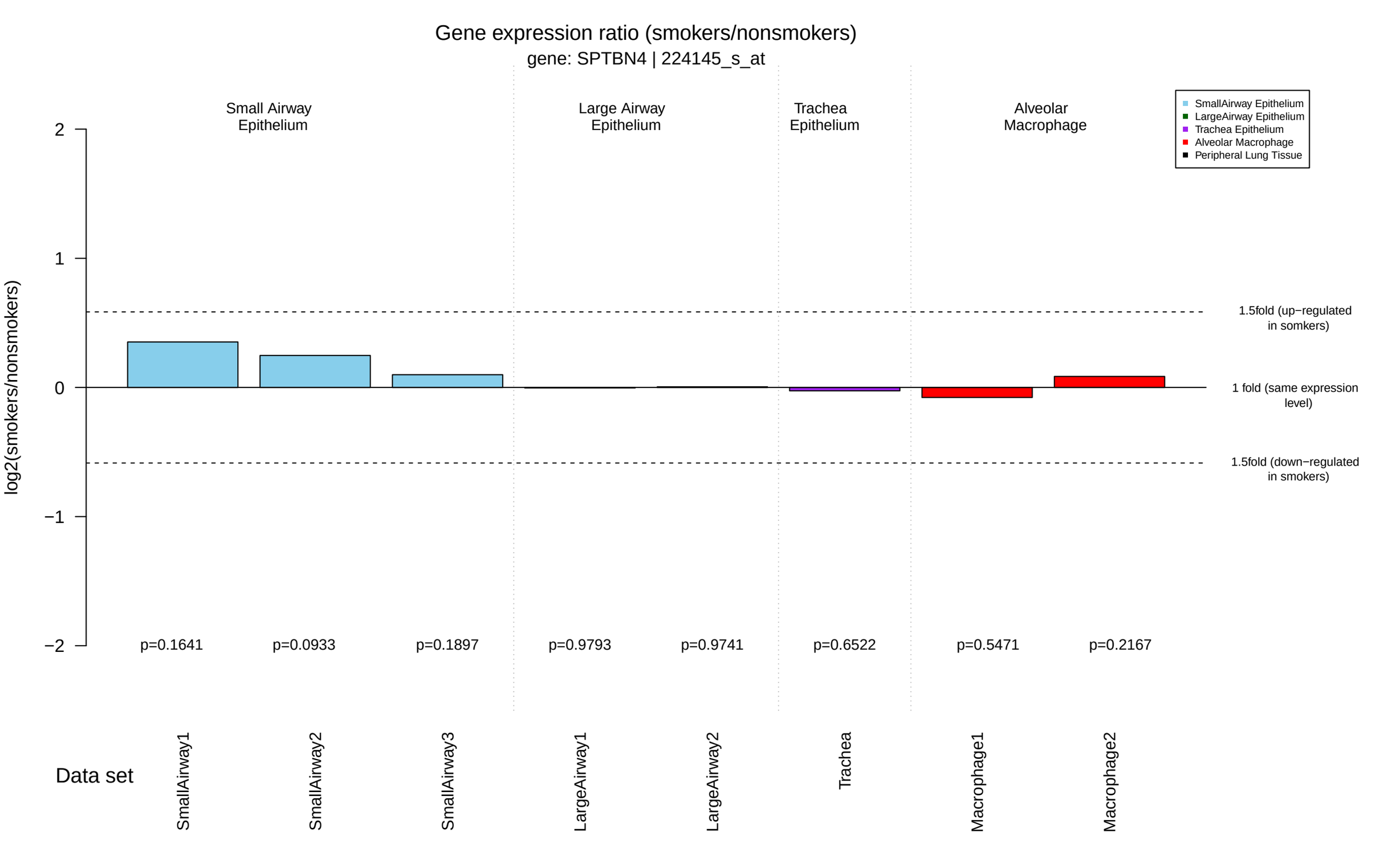
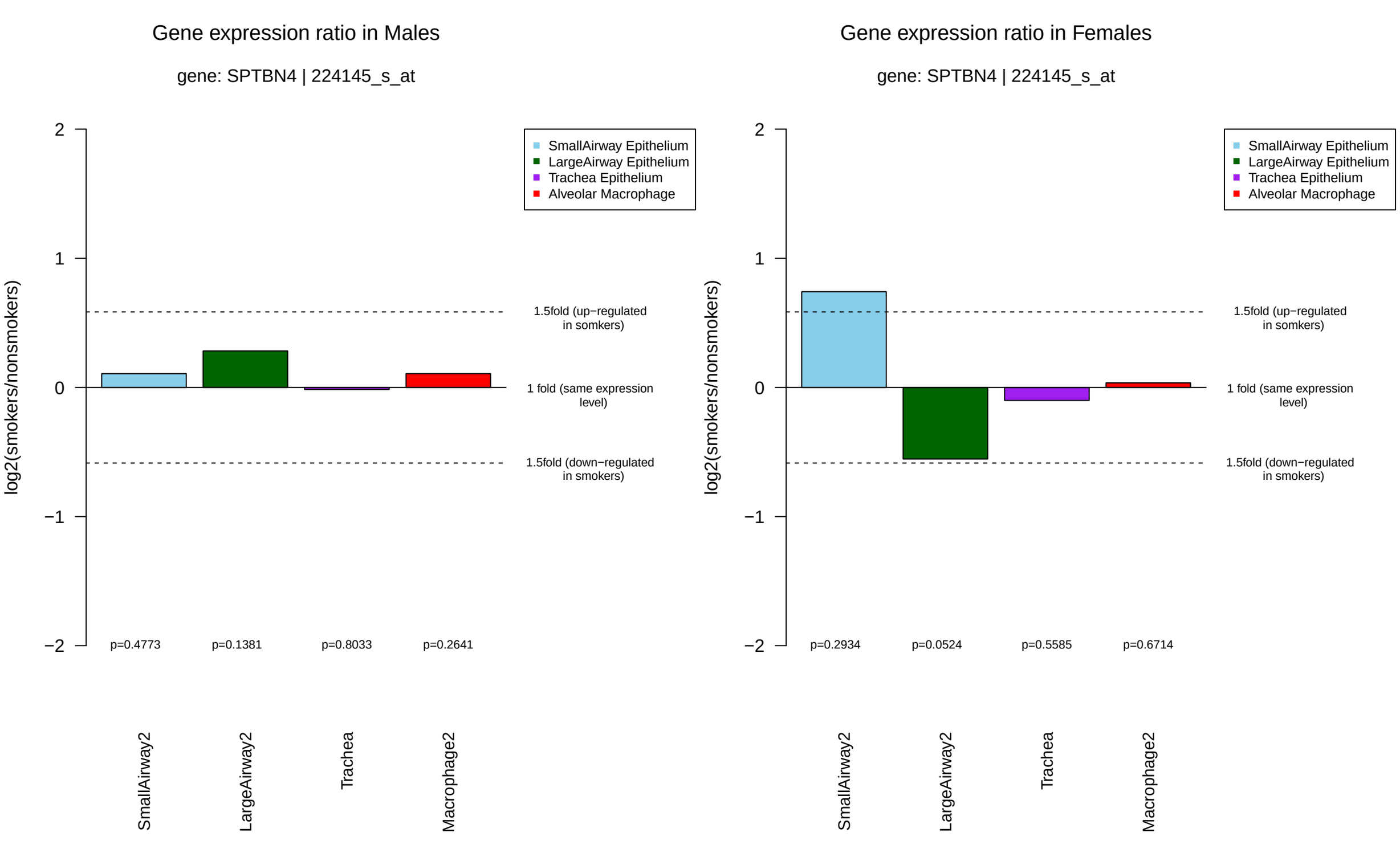
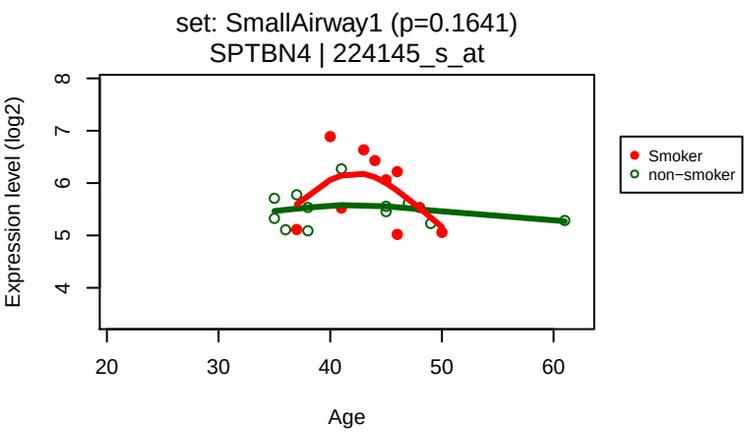
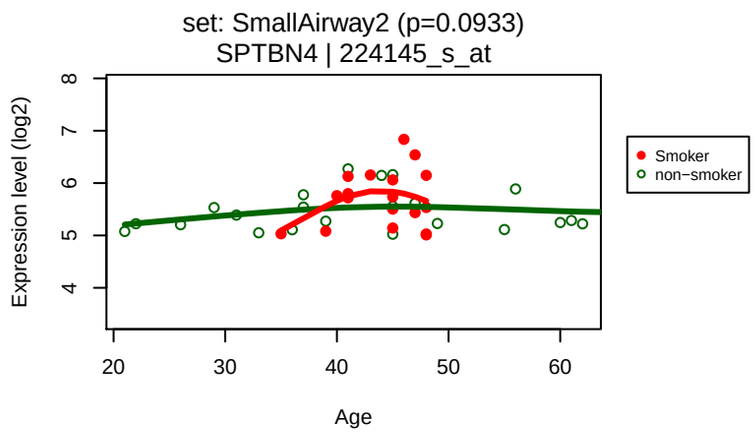
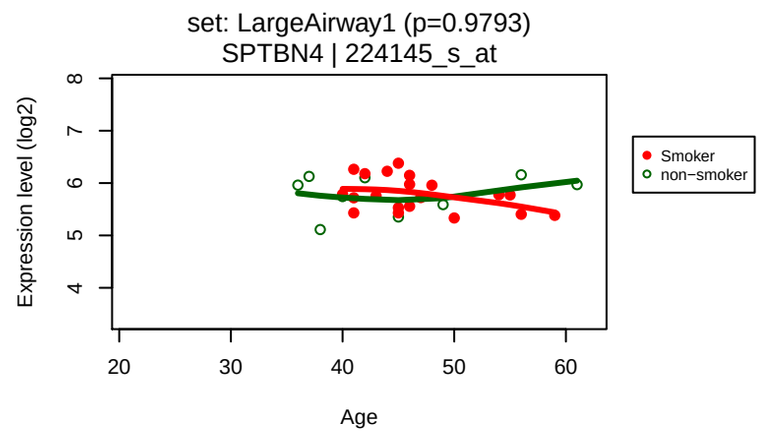

Gene expression of smokers and nonsmokers
gene: SPTBN4 | 224145_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1641
p=0.0933
p=0.1897
p=0.9793
p=0.9741
p=0.6522
p=0.5471
p=0.2167
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

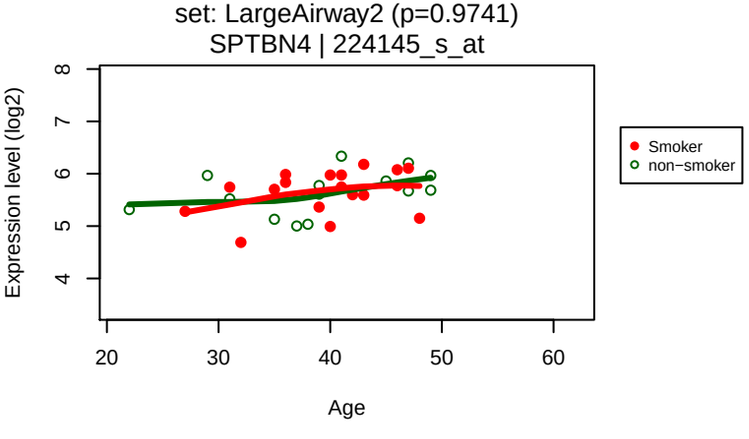

Gene expression ratio (smokers/nonsmokers)
gene: SPTBN4 | 224145_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1641
p=0.0933
p=0.1897
p=0.9793
p=0.9741
p=0.6522
p=0.5471
p=0.2167
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

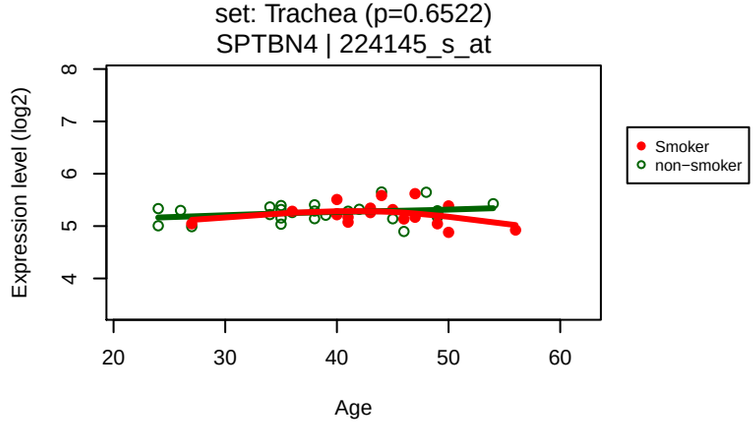

Gene expression ratio in Males
gene: SPTBN4 | 224145_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4773
p=0.1381
p=0.8033
p=0.2641
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SPTBN4 | 224145_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2934
p=0.0524
p=0.5585
p=0.6714
SmallAirway2
LargeAirway2
Trachea
Macrophage2

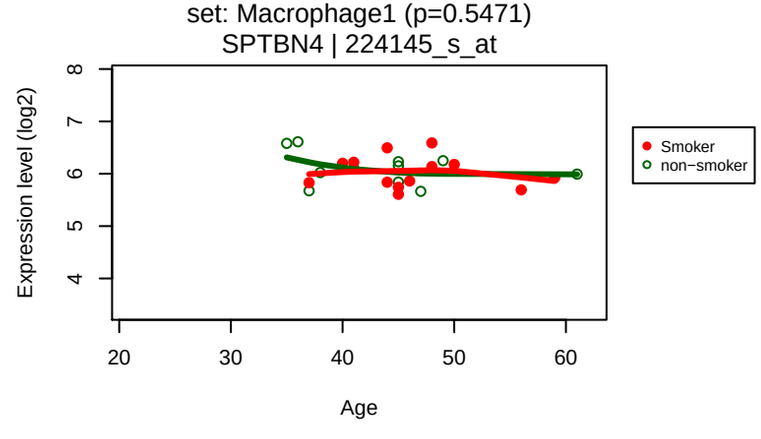

set: SmallAirway1 (p=0.1641)
SPTBN4 | 224145_s_at
Expression level (log2)
Age
Smoker
non−smoker

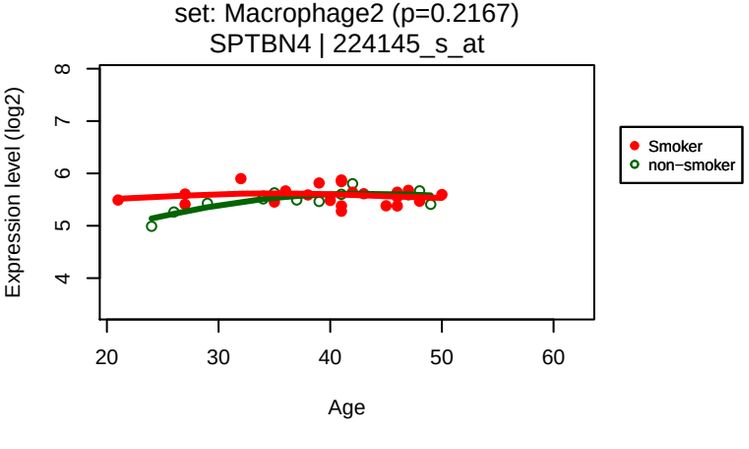

set: SmallAirway2 (p=0.0933)
SPTBN4 | 224145_s_at
Expression level (log2)
Age
Smoker
non−smoker

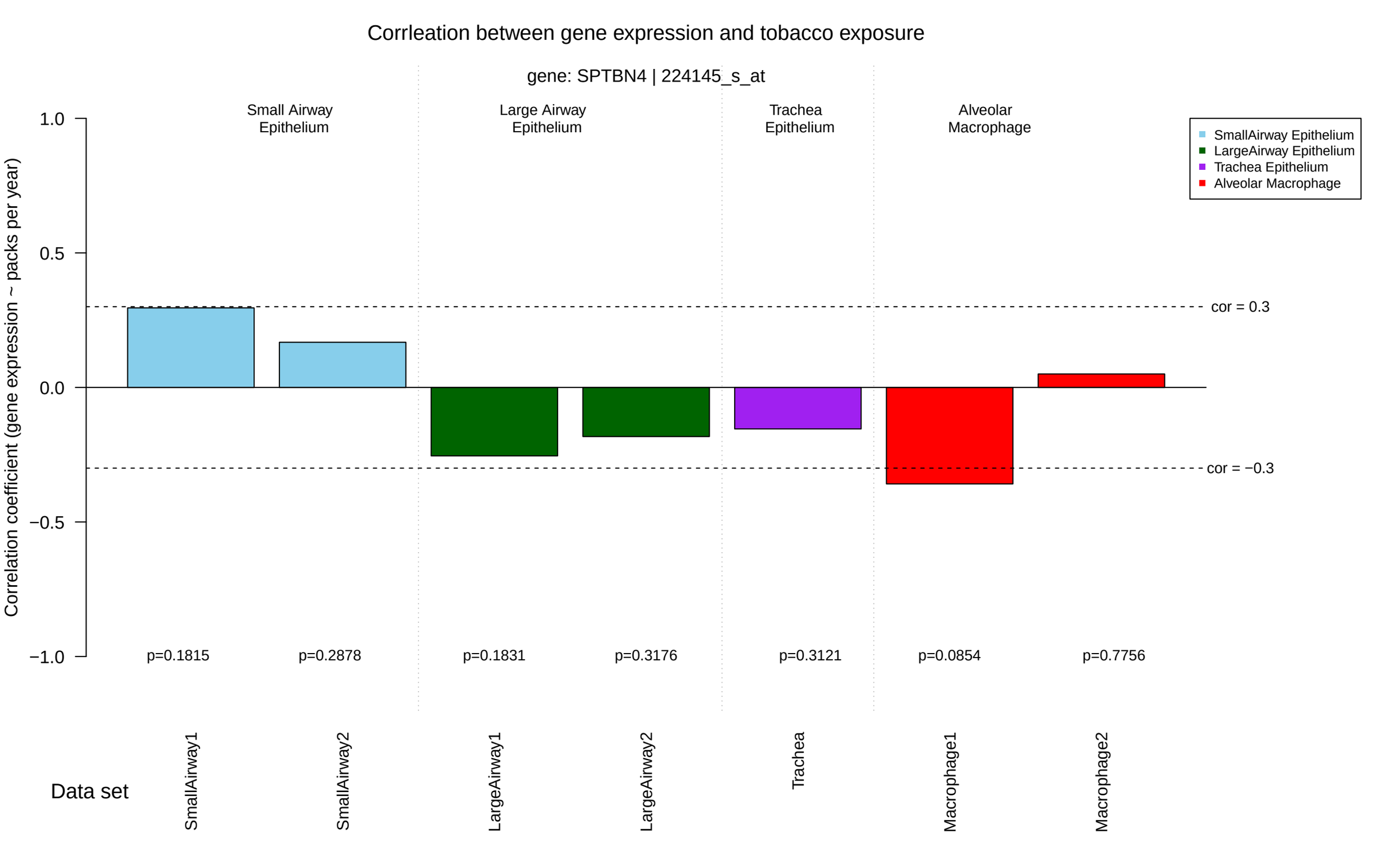

set: LargeAirway1 (p=0.9793)
SPTBN4 | 224145_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9741)
SPTBN4 | 224145_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6522)
SPTBN4 | 224145_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5471)
SPTBN4 | 224145_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2167)
SPTBN4 | 224145_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SPTBN4 | 224145_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1815
p=0.2878
p=0.1831
p=0.3176
p=0.3121
p=0.0854
p=0.7756
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2